Using Hailo computation power with Raspberry pi to run Deep learning models

We need to install all the Hailo dependency on both Raspberry pi and Ubuntu.

To install Hailo dependency on raspberry pi follow **Installing and Configuring the Hailo AI HAT.**

Setting up Hailo on Ubuntu

Install the Ubuntu 26 Version. (preferably dual boot than virtual-box, you are free to try virtual-box).

Install the following system requirements.(all are required)

Ubuntu® 22.04/24.04, 64-bit (supported also on Windows, under WSL2)
16+ GB RAM (32+ GB recommended)
Python 3.10/3.11/3.12, including `pip` and `virtualenv`
(Optional) For interactive visualization and graph support, install the following system packages:
`sudo apt-get install python3-tk graphviz libgraphviz-dev gcc pkg-config` Then install the visualization extra:
`pip install hailo-dataflow-compiler[viz]`
(Optional) Node.js (minimum version: v20.9.0) to start DFC Studio part of SDK package.

Create a virtual environment using Anaconda (you are free to use choice of your virtual env, I prefer anaconda).

Install Hailo Dataflow compiler and Hailo Model Zoo.(Check the Hailo developer webpage for latest and compatible Version.)



After installation, check by using the commands “hailo -h” and “hailomz -h”.

Creating Hailo Executable Format (.hef) from ONNX format

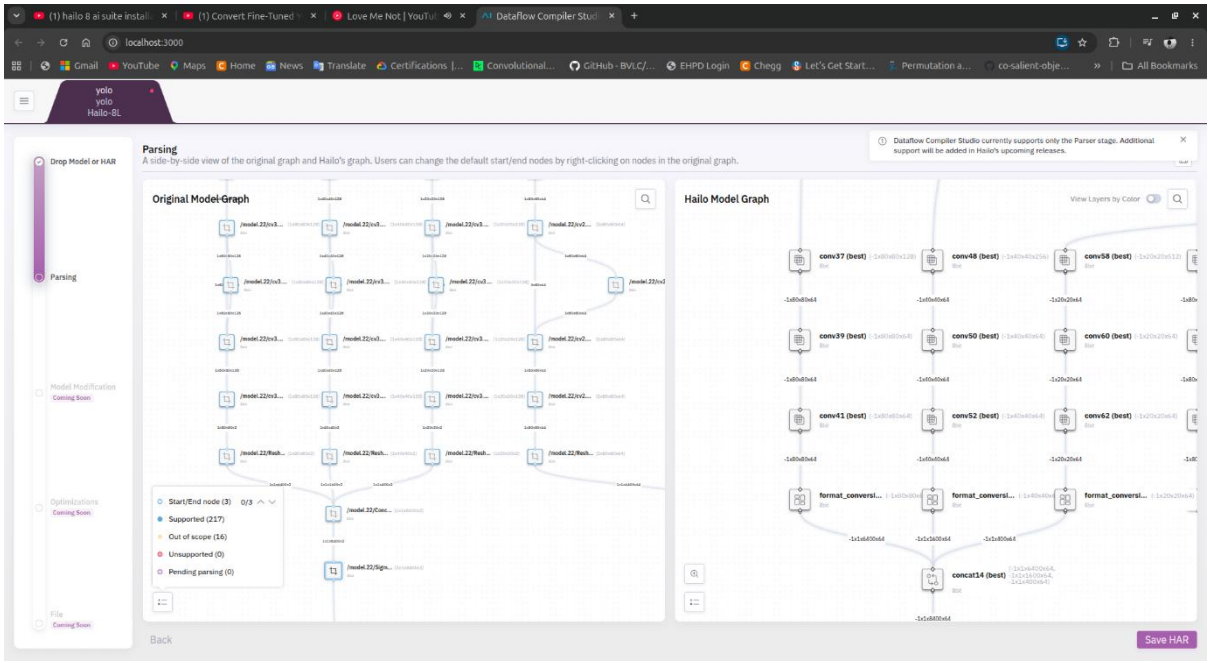
Get the ONNX model from the trained model, and few hundred validation(training) images from the dataset.

In ubuntu run hailo dfc-studio.

Open the local host given in the terminal and upload the ONNX file and parse the model.

It will generate the original model graph and Hailo model graph of the ONNX.

Make a note of starting node and the ending node from the graph.



Once it is done close and in terminal run the command

hailomz compile yolov8s

--ckpt best.onnx

--calib-path /run/media/satya-pallavarjula/SSD/personal/data/images/val

--hw-arch hailo8l

--classes 2

--start-node-names images

--end-node-names "/model.22/cv3.0/cv3.0.2/Conv" "/model.22/cv3.1/cv3.1.2/Conv" "/model.22/cv3.2/cv3.2.2/Conv" "/model.22/cv2.0/cv2.0.2/Conv" "/model.22/cv2.1/cv2.1.2/Conv" "/model.22/cv2.2/cv2.2.2/Conv"

Note change accordingly to your onnx model and hailo version.

This will generate a .hef